

ALI ABDULRAHMAN AHMED HASAN

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Multi-disciplinary Software Engineer and PhD Researcher holding both a Master's and Bachelor's in Software Engineering from Northeastern University. Born in Saudi Arabia with native Arabic, fluent English, and working Chinese proficiency, I bring a unique global perspective and deep technical alignment for enterprise data products. I bridge the gap between pure research and full-scale production by architecting secure, responsive web platforms, microservices, and big data pipelines from scratch using React.js, Vue.js, Node.js, and FastAPI. In tandem, I design and deploy end-to-end Machine Learning (ML) and Deep Learning pipelines using Python (TensorFlow, Keras, Scikit-learn, XGBoost) to extract actionable business insights from large-scale, unstructured datasets. My advanced background spans standard software engineering lifecycle deployments, hybrid databases (SQL, MongoDB, Redis), data distribution drift (Kolmogorov-Smirnov gating), and model performance calibration (Brier scores).

Work Experience & Research experience

PHD RESEARCHER & AI DEVELOPER, Northeastern University

SEPT 2024 — PRESENT, Shenyang, China

- Engineer predictive pipelines to intercept and measure live accuracy degradation in ensemble architectures (XGBoost) for financial modeling.
- Build statistical evaluation scripts in Python utilizing Kolmogorov-Smirnov (KS) tests to flag real-time structural data drift.
- Architect Python-based static analysis classes to comparative-mine software metrics (Cyclomatic Complexity, maintainability index) between AI-generated and human-authored syntaxes.
- Investigate computational intelligence gating frameworks to optimize cyberspace security and system reliability.
- End-to-End Data Preprocessing & Cleansing: Built automated Python pipelines (Pandas, Scikit-Learn) to cleanse, structure, and feature-engineer raw multi-dimensional datasets for downstream neural network training.
- Hybrid Database Architecture (SQL + NoSQL): Orchestrated dual-database environments leveraging MySQL for structured transactional integrity and MongoDB for high-throughput, unstructured social data streaming.

FREELANCE FULL-STACK DEVELOPER, Remote / Online Platforms

JAN 2021 — PRESENT, Online / Global

- Developed, deployed, and maintained custom web applications for multiple global clients using React.js, Vue.js, PHP (Laravel), and Node.js.
- Translated visual UI/UX mockups into pixel-perfect, responsive web interfaces, improving client-side rendering speeds and SEO metrics.
- Integrated third-party RESTful APIs, secure payment gateways, and authentication protocols (OAuth/JWT) into client applications.
- Successfully communicated technical requirements to non-technical business stakeholders, scoping project budgets, timelines, and deliverables independently.

FULL-STACK SOFTWARE ENGINEER (Trainee), Neusoft Education (Neuedu)

MARCH 2023 — MAY 2023, Shenyang, China

- Deployed component-driven, responsive tracking dashboards utilizing React.js and Vue.js connected to automated Node.js and Java API pipelines.
 - Managed visual Quality, Cost, and Delivery (QCD) reporting frameworks to proactively neutralize deployment risks and scope creep.
 - Collaborated within an enterprise engineering team to design and deploy scalable RESTful APIs and microservices using Java (Spring Boot) and Hibernate. Developed automated backend data pipelines, performing complex SQL database queries (Oracle/PostgreSQL) to optimize server-side application performance, reduce query latency, and ensure secure data transmission.
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Education

PH.D. CANDIDATE IN SOFTWARE ENGINEERING, Northeastern University
EXPECTED 2028, Shenyang, China

- Current Research GPA: 3.00

M.S. SOFTWARE ENGINEERING, Northeastern University
JULY 2024, Shenyang, China

- GPA:3.66

B.S. SOFTWARE ENGINEERING, Northeastern University
JULY 2021, Shenyang, China

Relevant Skills

React.js, Vue.js, Next.js, TypeScript, Python (FastAPI, Django), Java (Spring Boot, Spring Security, Hibernate, JPA) PHP (Laravel), Node.js, C/C++, (MySQL, PostgreSQL), NoSQL (MongoDB, Redis), Data Warehousing (Hadoop, Spark), Docker, Git CI/CD, Cyber Security, Agile/Scrum, CMMI Process Metrics, Native Arabic, Fluent English, Basic Chinese, Full Stack Developer , TailwindCSS , SaaS Multi-Tenant Machine Learning, Deep Learning, Predictive Modeling, NLP, Recommendation Engines, Clustering & Classification , Statistical Gating, Distribution Drift Analysis, Performance Calibration, Brier Scoring, Hypothesis Testing , Python, TensorFlow, Keras, PyTorch, Pandas, NumPy, Scikit-Learn, SciPy , Object-Oriented Programming .